The Shape of Wisdom: How Language Models Converge on an Answer

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Abstract

When a language model answers a factual question, where does its internal state go? We asked four open models (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Llama-3.2-1B) 2,256 factual questions whose answers recur, such as the country a person is a citizen of. At every layer, we recorded the state from which the answer is produced. Questions that receive the same answer converge on a shared region of the space of states, which we call the answer's *place*. The answer becomes readable from the state suddenly: in every model, readability jumps within a single layer, at a depth between 47% and 71% of the way through the network. A place does not depend on how the question is asked. For example, the place of a country learned from questions about people's citizenship also identifies that country when the question is about a town or a film. Places are also used by the model. After the jump, moving the state of a wrongly answered question into the right answer's place makes the model give the right answer in 83%–97% of cases late in the network, against 2%–4% for a random move of the same size. Before the jump, such moves do nothing. Finally, places reveal which mistakes are near misses. When a model answers wrongly but the right answer's place is close behind the place its state reached, the model usually picks the right answer when the question is asked as multiple choice. This geometry predicts such near misses better than the model's confidence does, in all four models.

1 Introduction

Ask a language model "What country is Aníbal Ponce a citizen of?" and, a few dozen layers later, it answers "Argentina". On the way, the model's internal state at the last position of the prompt, from which the answer is produced, changes at every layer. This paper asks what shape that change has. Does the state wander, drift slowly toward the answer, or arrive somewhere definite? And if it arrives somewhere, is that the same place for every question with the same answer?

We find that it is. As a question passes through the network, its state moves into a region that it shares with every other question to which the model gives the same answer. We call this region the answer's *place*. Figure 1 shows this in Llama-3.1-8B. Early in the network, questions whose answers are twelve different countries form one tangled cloud. Within a few layers, most of them separate into one strand per country.

We study places from four sides. First, we ask when the answer becomes readable from the state. In each of four models, readability jumps within a single layer, at a depth that differs between models (Section 3.1). Second, we ask whether a place depends on how the question is asked. It does not. The place of a country learned from questions about people also identifies that country in questions about towns and films, and it survives rewording (Section 3.2). Third, we ask whether the model uses places. Moving a state into a place after the jump makes the model give that place's answer. Random moves of the same size rarely do, and moves made before the jump have no effect (Section 3.3).

Fourth, we look at wrong answers. A place is defined by the answer the model gives, not by the true answer, so a wrong answer also lands in a place. Some wrong answers are near misses: shown the options, the model picks the right one. Others are real gaps in knowledge. The model's confidence barely tells the two apart, but where the state sits does. In a near miss, the right answer's place is close behind the place the state reached (Section 3.4).

In these models and on these questions, the answer does not sharpen steadily layer by layer. It appears at one depth, in a region shared with every other question that gets the same answer. A confident mistake reaches the region of the answer it gives in the same way.

2 Methods

In outline: we ask each model many factual questions whose answers recur, record its internal state at every layer, and compare the states of questions that receive the same answer. This section describes the questions (2.1–2.2), what we record (2.3), how we keep fitting and testing apart (2.4), and the three quantities used throughout: answer places, readability and tightness (2.5–2.7).

2.1 Questions

To see whether questions with the same answer reach the same state, we need many different questions for each answer. We built such a set from Wikidata [36]. It has 2,256 facts and 62 possible answers of four kinds: 24 countries, 16 cities, 12 languages[1] and 10 sports.

[1]We dropped Hindi as the language a person spoke after a manual check found its Wikidata labels unreliable. Hindi remains as the language of films.

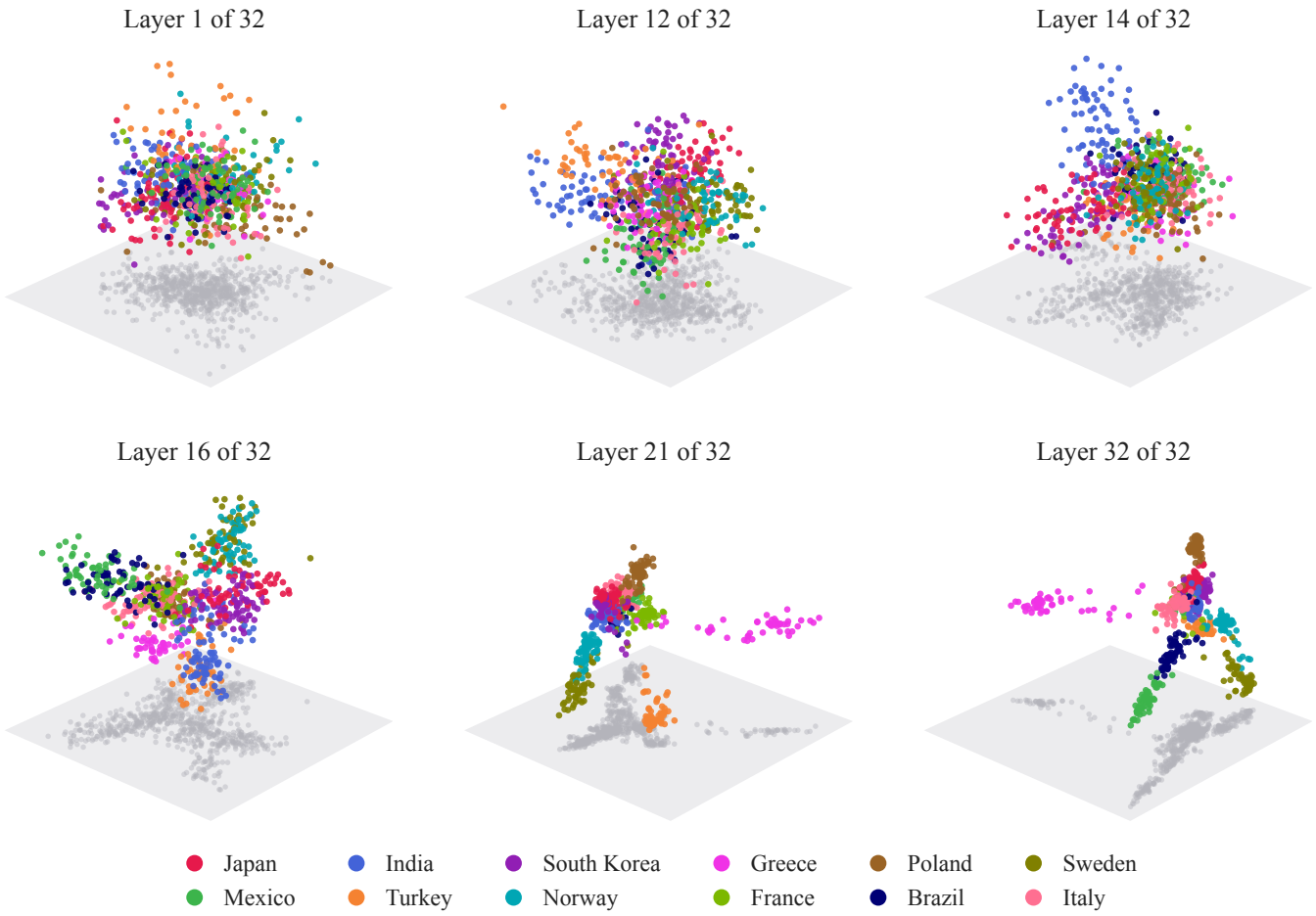


Figure 1: States of questions at six layers of Llama-3.1-8B. Each point is one test question about a person, town or film whose answer is one of the twelve countries the model gave most often; colour is the country the model answered. Each panel shows the questions' states at one layer, viewed along the three directions that best separate the twelve countries among the training questions at that layer (found by linear discriminant analysis). At layer 1 all questions share one cloud. At layers 12 and 14 a few countries, such as India, already stand apart. The onset layer (Section 3.1), at which answers become readable, is layer 15. By layer 16 most countries have begun to separate, and at the last layer most form a strand of their own. Each panel is scaled separately; grey dots are shadows on the floor.

Answer	Relation	Plain question
Country	Citizenship	What country is X a citizen of?
Country	Town's country	In which country is X?
Country	Film's country	Which country is the film X from?
City	Birthplace	In which city was X born?
City	Headquarters	In which city is X headquartered?
Language	Language spoken	Which language did X speak or write?
Language	Film's language	What is the original language of the film X?
Sport	Sport played	Which sport does X play?

Table 1: The eight relations. X is the subject: a person, town, film, company or athlete.

The facts come in eight types, listed in Table 1. Following Wikidata, we call a type of fact a *relation*. Several relations share a kind of answer. For example, a country can be the answer to a question about a person's citizenship, about where a town is, or about where a film is from. This lets us test later whether the same answer is reached in the same way from different relations.

Models know popular facts far better than rare ones [22], so we sampled subjects from three popularity bands, measured by the number of Wikipedia language editions (1–4, 5–19, and 20 or more), roughly a third from each. The models therefore know some facts and not others. Every subject has a single answer in Wikidata, a unique English name and no namesake of the same kind.

2.2 Wordings and formats

Each fact is asked as an open question in up to three wordings. The first is the plain question in Table 1. The other two are rewordings in assigned styles, such as a casual text message or a quiz host's question: "Hi, I'd appreciate knowing which country Aníbal Ponce is a citizen of, if possible?" A language model wrote the rewordings. It was told the subject and what to ask, but never the answer. We kept a rewording only if it named the subject exactly, asked a question, and mentioned no candidate answer; 4,323 rewordings passed.

The plain question is also asked as multiple choice, with four options shown in all four rotations of their order. The multiple-choice prompts are used only in Section 3.4, to tell

near misses from gaps; all other measurements use the open questions. In total, each model answers 15,603 prompts, each in the model's own chat format.

2.3 Models and states

We study four instruction-tuned models: Llama-3.1-8B-Instruct [12], Qwen2.5-7B-Instruct [32], Gemma-2-9B-it [7] and Llama-3.2-1B-Instruct. Each answers greedily.

A transformer keeps one vector per token position, the residual stream. Each layer reads this vector and adds its output to it. The answer is produced from the last position of the prompt, so we record the residual stream at that position: once before the first layer and once after every layer (before the final normalisation). We call this vector the question's *state* at that layer. It is a list of numbers, 4,096 of them in Llama-3.1-8B. *Depth* is the fraction of the model's layers that the state has passed through, from 0 at the input to 1 at the output. When we report a result at a given depth, we use the nearest layer.

We grade a reply by its first line. If that line names exactly one of the 62 candidates, matched against hand-curated names such as "China" or "Mandarin", the reply is right or wrong. A single answer outside the candidate list is wrong. Replies that hedge, refuse or name two answers are set aside (0.2%–8.7% of open replies, depending on the model).

2.4 Training and test questions

Several measurements first fit something to questions and then test it on other questions. To keep the tests fair, we split the subjects at random into two halves, once, and use the same split everywhere. Questions about the first half are *training questions*; questions about the second half are *test questions*. Everything that is fitted uses training questions only. Every number we report is measured on test questions. We split by subject, not by question, because the questions about one subject (its wordings and formats) are near copies of each other.

2.5 Answer places

An answer's *place* at a layer is the average state, at that layer, of the training questions to which the model gave that answer:

$$p_a = \frac{1}{|Q_a|} \sum_{q \in Q_a} h_q, \tag{1}$$

where

- a is an answer, for example Argentina;
- Q_a is the set of training questions that the model answered with a, and $|Q_a|$ is their number;
- h_q is the state of question q at the layer;
- p_a is the place of a at the layer.

For example, the place of Argentina at layer 15 of Llama-3.1-8B is the average layer-15 state of all training questions that Llama answered with "Argentina", whether they asked about a person, a town or a film, in any wording. Each answer has one place per layer. We build a place only if the model gave that answer at least five times, for at least three different subjects.

Places are built from the answers the model gives, not from the true answers. If the model wrongly answers "Chile", the question counts toward the place of Chile. This lets us ask later where wrong answers sit.

2.6 Readability

Readability measures whether the answer a model gave can be identified from its state alone. For each test question, we find the place nearest to its state, among answers of the same kind:

$$\hat{a}_q = \arg\min_{a \in A_q} \big\| W(h_q) - W(p_a) \big\|, \tag{2}$$

where

- h_q is the state of test question q at the layer;
- A_q is the set of answers, with a place, of the same kind as the answer the model gave to q, for example all country answers for a country question;
- p_a is the place of answer a at the same layer;
- W is a whitening, described below, applied to both;
- $\|\cdot\|$ is the ordinary straight-line distance between two vectors;
- $\hat{a}_q$ is the answer whose place is nearest.

Readability at a layer is the share of test questions for which $\hat{a}_q$ is the answer the model actually gave. For example, a readability of 70% means that for 70% of test questions, the nearest place belongs to the model's own answer. We count only test questions whose given answer has a place. If states carried no information about answers, readability would be about one over the number of places, for example 4% for the 24 countries.

Whitening is needed because a few dimensions of the residual stream take much larger values than the rest, and would decide any distance on their own. Whitening makes every direction count equally. At each layer, using training questions only, we (i) subtract the average state, (ii) find the 128 directions along which states vary most (their principal components), (iii) describe each state by its 128 coordinates along these directions, and (iv) divide each coordinate by the spread of training states along that direction.[2]

2.7 Tightness

Readability depends on the whitening. As a check that does not, we also measure tightness in the model's own, unwhitened states. Tightness compares how far questions sit from their own place with how far apart the places are:

$$T = \frac{S_{\text{within}}}{S_{\text{between}}}, \tag{3}$$

[2]Each spread is first shrunk 10% toward the average spread, so that directions with very little spread are not blown up.

with

$$S_{\text{within}} = \frac{1}{N} \sum_{q} \|h_q - p_{g_q}\|^2,$$

$$S_{\text{between}} = \frac{1}{|A|} \sum_{a \in A} \|p_a - \bar{p}\|^2,$$

where

- the sum in S_{within} runs over the N test questions whose given answer has a place;
- g_q is the answer the model gave to question q, so p_{g_q} is that answer's place;
- S_{within} is therefore the average squared distance of a question from its own place: how widely questions scatter around their places;
- A is the set of answers that have a place, and $|A|$ is their number;
- $\bar{p}$ is the average of all these places;
- S_{between} is therefore the average squared distance of a place from the centre of all places: how far apart the places are.

A small T means that questions gather tightly around places that are far apart. A large T means that questions scatter far more widely than the places are separated, so places cannot be told apart. We compute S_{within} and S_{between} for each kind of answer separately, and add each over the kinds, weighting each kind by its number of test questions.

2.8 Uncertainty and controls

Questions about the same subject are not independent: a model tends to get all of them right or all of them wrong. We therefore estimate uncertainty by resampling subjects. We draw the test subjects at random, with replacement, 300 times, recompute the result each time, and report the range that contains the middle 95% of the results.

Two controls show how much readability could appear without any information about the answer in the states:

- *Shuffled answers.* We randomly reassign the given answers among facts of the same kind, then build places and measure readability exactly as before (average of five shuffles). The states are unchanged, but their link to the answers is broken, so what remains is chance.
- *Question text only.* We replace each state with a vector made from the question's text alone, without the model: the counts of every sequence of two to four characters, weighted by how rare each sequence is (TF-IDF) and compressed to 256 numbers by a truncated singular value decomposition fitted on training questions. This shows how much of the answer could be guessed from the wording, for example from the spelling of a name.

3 Results

We report five results: when the answer becomes readable (3.1), whether places are shared across different questions (3.2), whether the model uses places (3.3), what places reveal about wrong answers (3.4), and whether places follow the given or the true answer (3.5). Table 2 collects the main numbers for each model.

3.1 When the answer becomes readable

Figure 2A shows readability layer by layer. In every model the curve has the same shape: a low plateau, a jump, and a high plateau. The jump takes a single layer. In Llama-3.1-8B, readability rises from 32% at layer 14 to 71% at layer 15. The other three models show the same one-layer jump (Table 2). After the jump, readability keeps rising to a peak of 86%–94%.

We call the layer of the jump the *onset layer*, and its depth the *onset depth*. The onset depth differs between models, from 47% to 71% of the way through the network. We find the onset layer without looking at test questions: we split the training subjects in half again, build places from one half, measure readability on the other, and take the first layer at which this readability covers half the distance from chance to its highest value over layers. Chance here is the average, over questions, of one over the number of places of the question's kind.

Both controls stay low at every layer, in every model: 6%–8% with shuffled answers and 11%–18% with the question text only. Before the jump, readability is 20%–55%, above both controls. This early signal may come from surface cues in the subject's name, such as its language; we did not test this. It does not grow toward the jump. The jump is smaller for sports, the one kind of answer that is not a name (Section 5).

Tightness shows the same event without whitening (Figure 3). From its highest to its lowest value, it falls 16- to 25-fold, and the fall is concentrated at the onset. Two of each model's 16 to 42 layers, the one into the onset layer and the one out of it, carry 16%–32% of the whole fall (on a log scale). Tightness then keeps falling more slowly for about ten layers. This differs from trained classifiers, in which the classes separate by a roughly constant factor at every layer [17].

3.2 Places are shared across questions

If a place stands for the answer itself, it should not depend on what exactly the question asks or how. We test this in three ways, each a change to the standard readability measurement (Figure 2B, Table 2). In each test, a drop in readability would mean that places depend on the thing changed.

Other relations. We build places from the training questions of one relation only, for example citizenship, and measure readability on test questions of the other relations with the same kind of answer, for example the countries of towns and films. We do this for each relation in turn and average the results. Sports, with a single relation, are left out of this test.

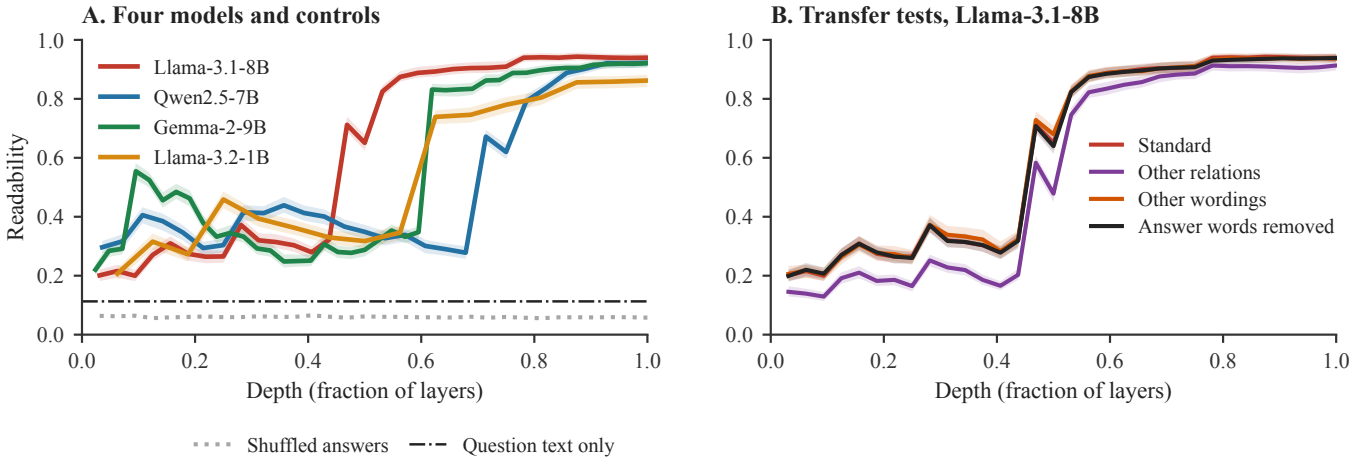


Figure 2: Readability by depth. (A) All four models, with the two controls of Section 2.8 (shown for Llama-3.1-8B). (B) Llama-3.1-8B under the three tests of Section 3.2; the standard, other-wording and answer-words-removed curves nearly coincide. Bands are 95% intervals.

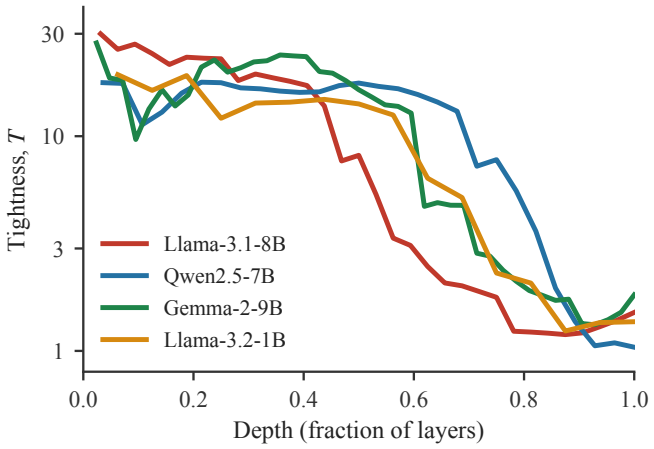


Figure 3: Tightness by depth (Equation 3, log scale). Lower is tighter.

	Llama 8B	Qwen 7B	Gemma 9B	Llama 1B
Layers	32	28	42	16
Accuracy	63%	58%	67%	48%
Onset layer	15	20	26	10
Onset depth	47%	71%	62%	62%
Readability				
Layer before onset	32%	28%	35%	35%
Onset layer	71%	67%	83%	74%
Peak	94%	92%	92%	86%
Standard, 90% depth	94%	90%	92%	86%
True-answer places	75%	65%	73%	61%
Other relations	91%	85%	90%	76%
Other wordings	94%	90%	92%	85%
Answer words removed	94%	90%	91%	85%
Telling near misses from gaps (AUC)				
Rank of right place	0.77	0.75	0.75	0.82
Confidence	0.66	0.53	0.49	0.66

Table 2: Per-model summary. Accuracy: share of plain test questions answered correctly. True-answer places: Section 3.5. The last three readability rows are the tests of Section 3.2. All readability rows from "Standard" down are at 90% of depth. AUC: Section 3.4.

In Llama-3.1-8B, readability at 90% of depth is 91%, against 94% in the standard measurement on the same kinds of answer, in which places are built from all relations. The place of a country is therefore much the same whether the question is about a person, a town or a film.

Other wordings. We build places from plain questions only and measure readability on the rewordings. Readability is 94%, the same as in the standard measurement.

Answer words removed. A place could simply be the model preparing to say a particular word, since that would also be shared across questions. To test this, we removed from every state the directions through which the model writes the answer words. The model produces a word by comparing its final state with one fixed vector per token (its unembedding vector), after a normalisation that rescales each dimension. We took these vectors, rescaled in the same way, for every token of the 62 answer names, and removed from each state every direction they span, keeping only the part of the state at right angles to them. Readability did not change (94%). Removing the same number of random directions did not change it either. Places can therefore be read without the directions that write the answer word.

3.3 The model uses places

Readability shows that the answer can be read from the state. It does not show that the model uses this information. To test that, we moved states into places and checked whether the answer changed. For each test question that the model had answered wrongly, with an answer that has a place (121–233 questions per model), we changed its state at one layer to

$$h' = h + (p_{\text{right}} - p_{\text{given}}) , \tag{4}$$

where

- h is the question's state at that layer, at the last prompt position;
- p_{given} is the place of the wrong answer the model gave;

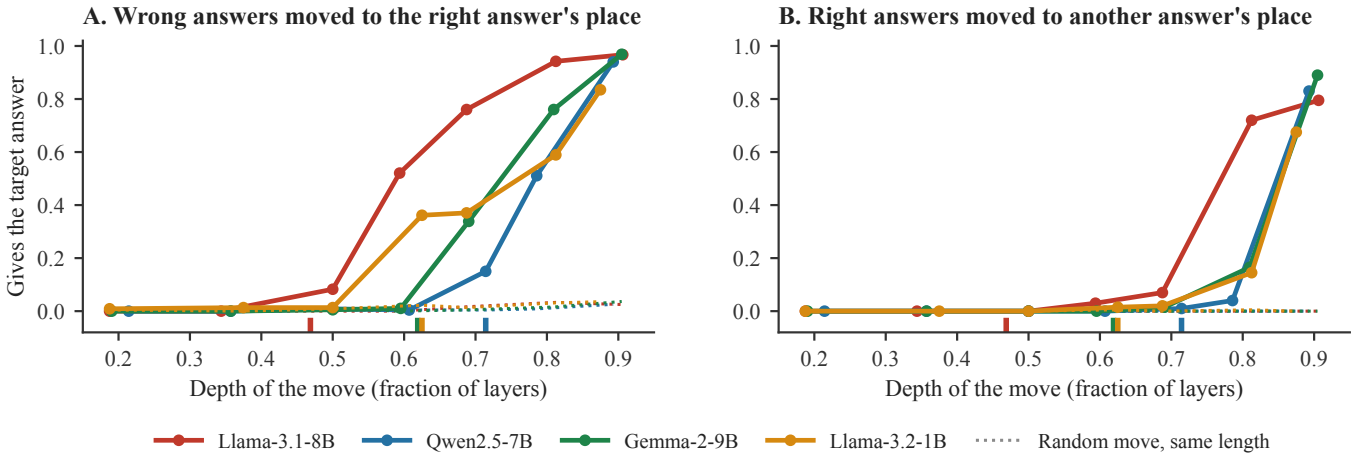


Figure 4: Moving states toward places (Section 3.3). (A) Wrong answers moved toward the right answer's place. (B) Right answers moved toward another answer's place. Solid lines: moves toward the place; dotted lines: random moves of the same length. Short bars under each axis mark each model's onset depth.

- p_{right} is the place of the right answer;
- h' is the moved state.

The move therefore shifts the state by exactly the gap between the two places. The move is made once, while the model reads the prompt; the model then generates its reply without further changes. We made this move at one depth at a time, at seven depths: 20%, 35%, 50%, 60%, 70%, 80% and 90% of the network. We also ran two controls: a move of the same length in a random direction, and a move toward the place of a third answer, neither given nor right, drawn at random from the places of the same kind. Finally, we took questions the model had answered correctly and moved them toward the place of another answer of the same kind, also drawn at random.

Moves work only after the onset (Figure 4). At 20% of depth they correct 0%–1% of wrong answers. After the onset, the later the move, the more often it works. At about 90% of depth it corrects 97% of wrong answers in Llama-3.1-8B, 94% in Qwen2.5-7B, 97% in Gemma-2-9B and 83% in Llama-3.2-1B. Random moves of the same length produce the right answer in only 2%–4% of cases.

The effect is not specific to the right answer. At 90% of depth, wrong answers moved toward a third answer's place give that answer in 82%–95% of cases, and right answers moved toward another answer's place switch to it in 68%–89% of cases.

One caveat concerns size. Because places move apart with depth, the moves grow from 3%–5% of the state's length at 20% of depth to 42%–49% at 90%. Late moves are large and close to the output, and may partly write the answer directly. The clearest evidence is therefore the depth at which moves start to work, and the failure of random moves of the same length.

3.4 Wrong answers: near misses and gaps

Because places are built from the answers the model gives, a wrong answer has a place too, and we can ask what the state of a wrong answer looks like. Some wrong answers are *near misses*: the model fails the open question but picks the right option when the same question is asked as multiple choice. We count a wrong answer as a near miss only if the model picks the right option in at least three of the four option orders. This holds for 6%–60% of wrong answers, fewest in Llama-3.2-1B. The other wrong answers we call *gaps*.

Can near misses be told apart from gaps? We compared two single numbers for each wrong answer to a plain test question whose right answer has a place:

- *Rank of the right place.* At the onset layer, where answers first become readable, we rank all places of the question's kind by their distance from the state (in whitened states), with 1 for the nearest, and take the rank of the right answer's place.
- *Confidence.* The model's average log-probability per token of its own reply.

We measure how well each number separates near misses from gaps with the AUC: the probability that a randomly chosen near miss scores better than a randomly chosen gap (a nearer right place, or a more confident reply). An AUC of 0.5 is chance and 1 is perfect separation. Both AUCs use the same wrong answers, and neither number is fitted to the data.

The rank of the right place separates near misses from gaps better than confidence in every model (Figure 5): AUC 0.75–0.82, against 0.49–0.66 for confidence. The 95% interval for the difference excludes zero in every model. For near misses, the right answer's place has a median rank of 1–2, depending on the model; for gaps it is 4–8. A near miss is thus visible inside the model: the right answer's place is among the nearest, often just behind the place of the answer the model gave. The model's confidence in its reply barely shows this.

3.5 Places follow the given answer

Places are built from the answers the model gives. Two results show that this is also what they represent, rather than the true answer. First, take two models, M and N, and the plain test

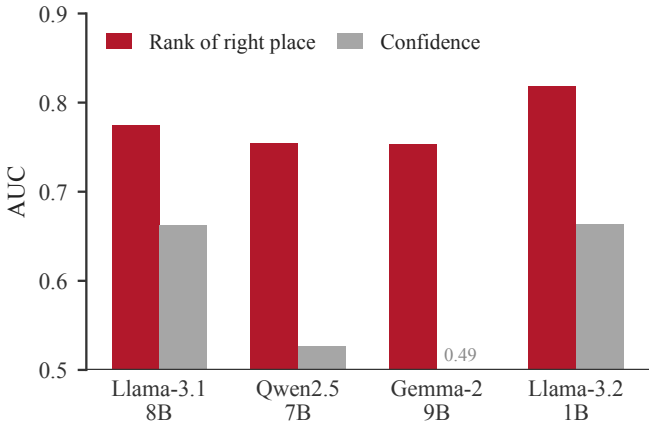


Figure 5: Telling near misses from gaps (Section 3.4). AUC of each single number among wrong answers; 0.5 is chance.

questions to which they give different answers. On these questions, M's state at its onset layer lies nearer M's place for M's answer than M's place for N's answer in 66%–88% of cases (the range over all 12 ordered pairs of models). Second, we built places from the true answers instead, and measured how often the nearest place is the true answer. At 90% of depth this readability is 61%–75%, well below the 86%–94% of places built from the given answers (Table 2).

4 Related work

Reading intermediate layers. The logit lens [26] decodes an intermediate state through the model's output layer, and the tuned lens [1] learns a decoder for each layer. Both show predictions sharpening with depth. Future Lens [28] and Patchscopes [10] read richer content from a single hidden state. All of these read the state through the output vocabulary. Our places are fitted from the states themselves, and reading them does not need the answer words' output directions.

How depth is used. Several studies find that layers play different roles. Early layers gather and detokenise, middle layers build abstract features, and late layers refine a prediction [3, 4, 13, 20, 35]. Correct and incorrect predictions follow similar confidence curves across layers [19]. In factual recall, information about the relation and then the subject moves to the last position, where attention heads extract the attribute [9]. Feed-forward layers act as key–value memories [8], and middle-layer feed-forward modules at the subject's last token are causally decisive [24]. Our onset depth marks, for a whole population of questions, the depth at which each model's answer becomes available at the last position.

Representation geometry. Middle layers represent the same content in a shared space across languages and modalities [37, 41]. Concepts have structured geometry. Categories form simplices [30], some features are circular [5], space and time are linearly decodable [14], counts lie on curved manifolds that attention reshapes [15], and truth has a linear direction [23]. In trained classifiers, representations of different classes separate by a roughly constant factor per layer [17]. Language models move partway toward neural collapse, in which states gather at their class means, at their top layer, and further with scale [40]. The residual stream contains stable regions, within which small changes leave the next-token prediction unchanged [18]. Yet, almost surely, no two prompts map to the same last-position state [25]. Places fit both findings: questions can gather in the directions that carry the answer while staying distinct in others.

What models know but do not say. Internal states often encode more than the output shows [6, 27, 33]. On knowledge benchmarks such as MMLU and ARC, however, this gap is near zero [31], and internal states mostly reflect what the model recalls rather than what is true [2]. False few-shot demonstrations can override a correct internal answer at a critical layer [16], and users' suggestions shift answers [34]. In multiple choice, attention in a few middle layers selects the answer letter, and a sparse set of heads then raises its probability [38]. The winning option is chosen by its content before it is bound to its letter [39], and models carry a prior over letters [42].

Steering. Adding a direction to the residual stream changes a model's behaviour [21, 29]. Our moves are of this kind. Each one, however, takes a single question's state from one measured place to another at one layer, and is compared with random moves of the same length.

5 Limitations

The questions are short and factual, and their answers come from a closed set of 62 countries, cities, languages and sports. Three of these four kinds are names. For the one kind that is not, sports, the rise at the onset layer is smaller (7–19 percentage points, against 41–52 for names), and readability at 90% of depth is lower (63%–90% against 89%–94%). Sports have at most ten places, all from one relation, so we cannot tell whether this reflects the kind of word or the smaller set. Places for open-ended answers, multi-step reasoning or long contexts may look different.

We studied four instruction-tuned models between 1 and 9 billion parameters, with one prompt format each and greedy answers. Readability and tightness both measure where states sit, not the computation that moves them there. Readability jumps within one layer, but tightness keeps falling for about ten layers after the onset, so the convergence itself is not instantaneous. Moves at late layers may partly write the answer directly, so the strongest causal evidence is the depth at which moves start to work and the failure of random moves.

Wikidata answers are a reference, not ground truth. We removed one unreliable relation–answer pair after a manual check, and a small number of wrong labels surely remain. The rewordings were written by a language model and filtered automatically.

6 Conclusion

We followed the state from which four language models answer short factual questions, layer by layer. In all four, the answer becomes readable from the state within a single layer. The depth of this onset differs between models. From the onset on, questions with the same answer gather in a shared place. The place does not depend on the relation or the wording of the question, and it can be read without the directions that write the answer's name. Moving a state into a place after the onset makes the model give that place's answer; before the onset, the same move does nothing. Places follow the answer the model gives rather than the true one, so wrong answers land in places too. Where they land separates near misses, which the model recognises when shown the options, from real gaps in knowledge, and it does so better than the model's confidence. Whether the same picture holds for open-ended answers, for reasoning over several steps and for larger models is the natural next question.

Code and data. Code, the dataset and all derived results: https://github.com/gut-puncture/The-Shape-of-Wisdom.

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